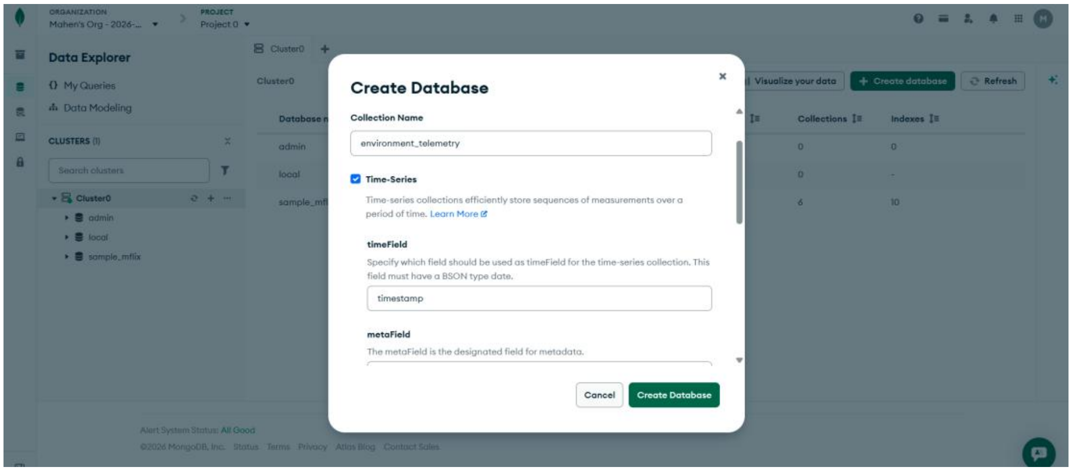
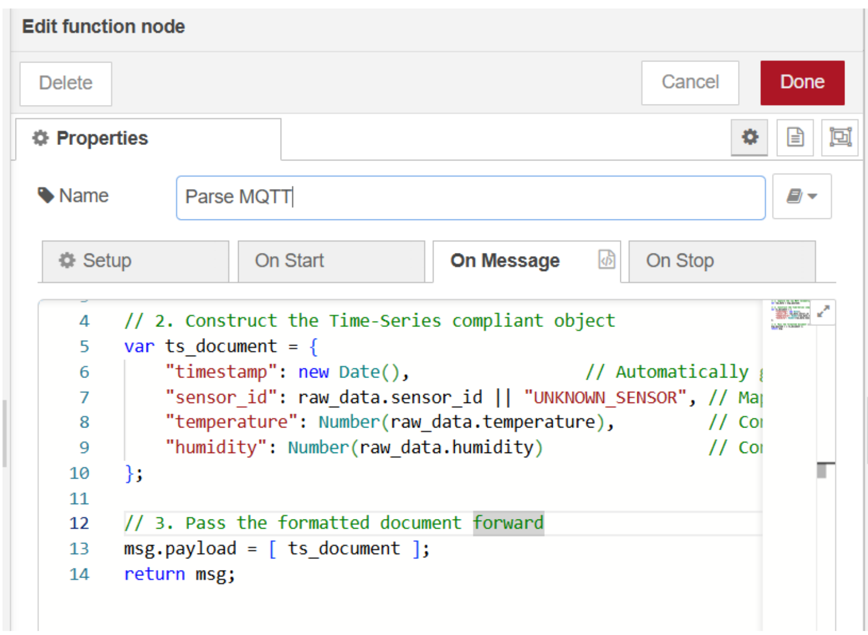
MAHENDRAN A/L THILAGAMANY B122510490

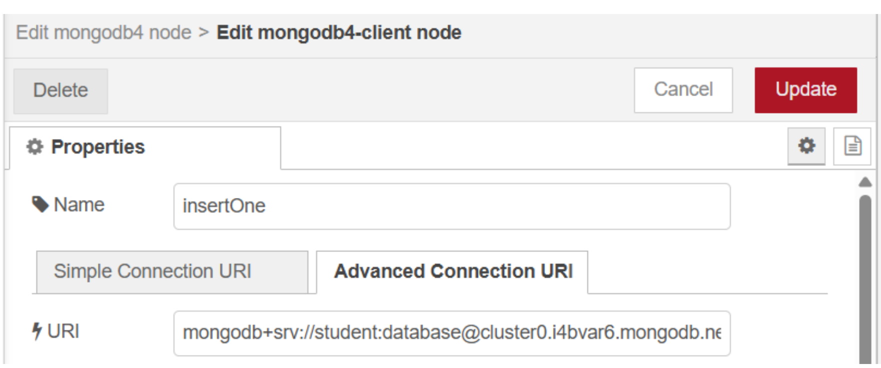
EXERCISE 8: TIME-SERIES ARCHITECTURE

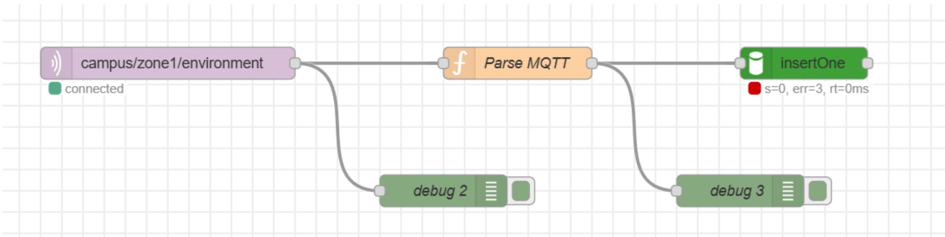
Step 1: Architecting the Time-Series Collection in Cloud

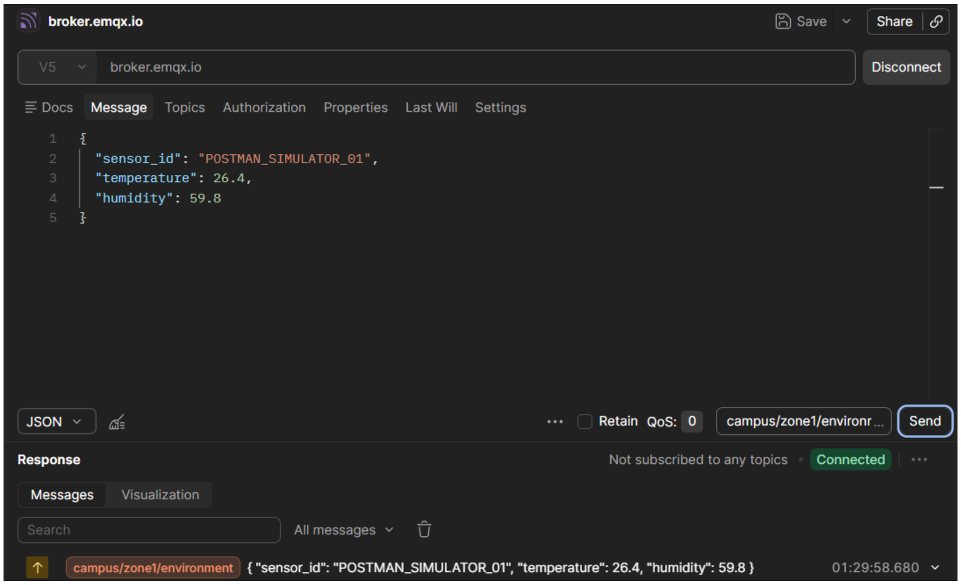


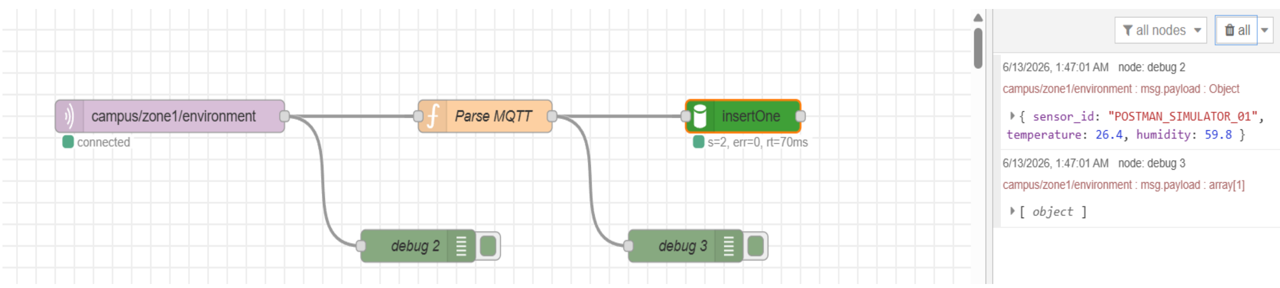
Step 2: Node-RED Middleware Transformation



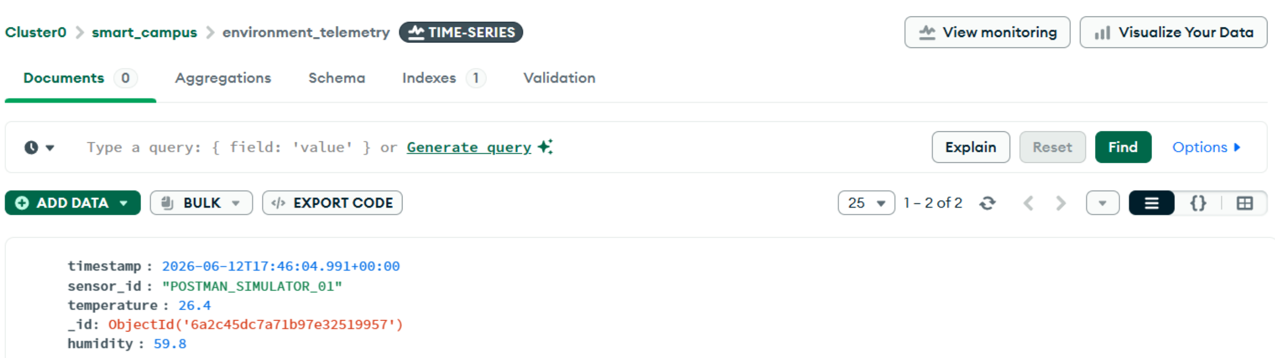








Step 3: Verifying Cloud Indexes and Bucket Storage



Part 4: Discussion & Architectural Analysis

Question 1: The Definition of Metadata

Why Metadata Must Be a Static Identifier

In a MongoDB Time-Series architecture, the metaField serves as the **uniquely identifying label** for a distinct data stream. It represents the source or the constant characteristics of the environment being measured (e.g., a specific device ID, MAC address, or deployment zone). It must be static because it acts as the primary key used to group data points into logical collections over time.

Impact on the "Bucket Pattern" Compression

MongoDB utilizes an internal architecture known as the **Bucket Pattern**. Instead of storing every incoming MQTT message as an isolated document on the disk, it automatically creates a compressed container (a bucket) for each unique metadata value over a specific time window.

Question 2: Middleware vs. Edge Timestamping

Why Server-Side Timestamping is Generally Safer

Injecting the timestamp inside the Node-RED middleware layer via new Date() ensures absolute chronological consistency and data integrity. It removes the reliance on edge hardware state, which is notoriously unpredictable in industrial or distributed campus environments.

- **Lack of native Real-Time Clocks (RTC):** A standard ESP32 does not have a built-in, battery-backed hardware clock. When it boots up, its internal time register starts at zero (1970-01-01 00:00:00 UNIX epoch).
- **Network & Synchronization Dependencies:** To get the correct local time, the ESP32 must successfully connect to the internet and query an external Network Time Protocol (NTP) server. If the campus Wi-Fi drops momentarily or the NTP handshake fails, the microcontroller will broadcast telemetry with inaccurate or blank dates.
- **Power Cycle & Battery Volatility:** If an edge sensor experiences a brownout, power spike, or battery replacement, its volatile memory resets completely.